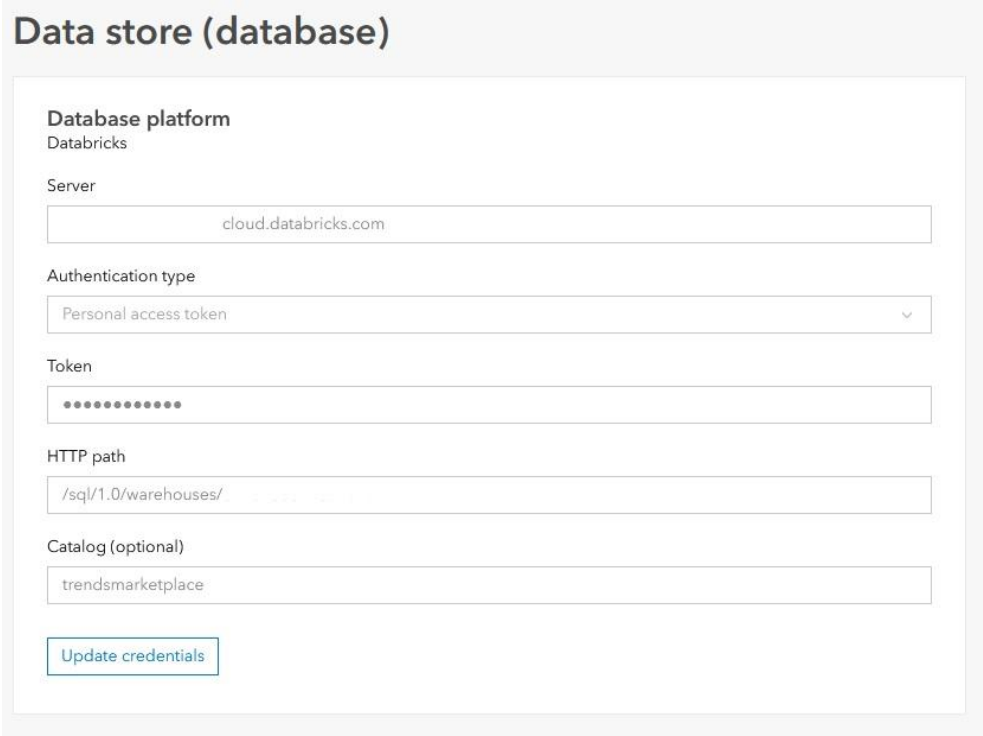
ArcGIS Data flow

Establishing Database connection between ArcGIS Online and Databricks cloud storage

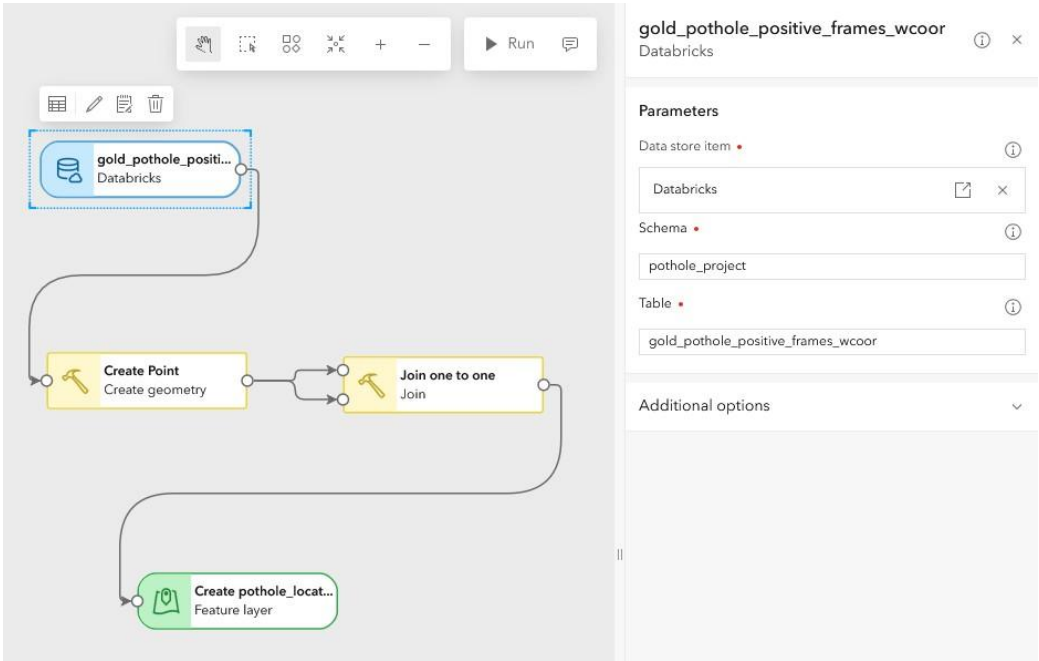
This is a persistent connection to the catalog in databricks that contains the model, labeled data, and unlabeled data. This connection will only be used to view a table containing predicted positive images. Each row will contain the directory address for the image, the predicted probability for that image, and the coordinates for that image. The expiration of the key is configurable in Databricks.



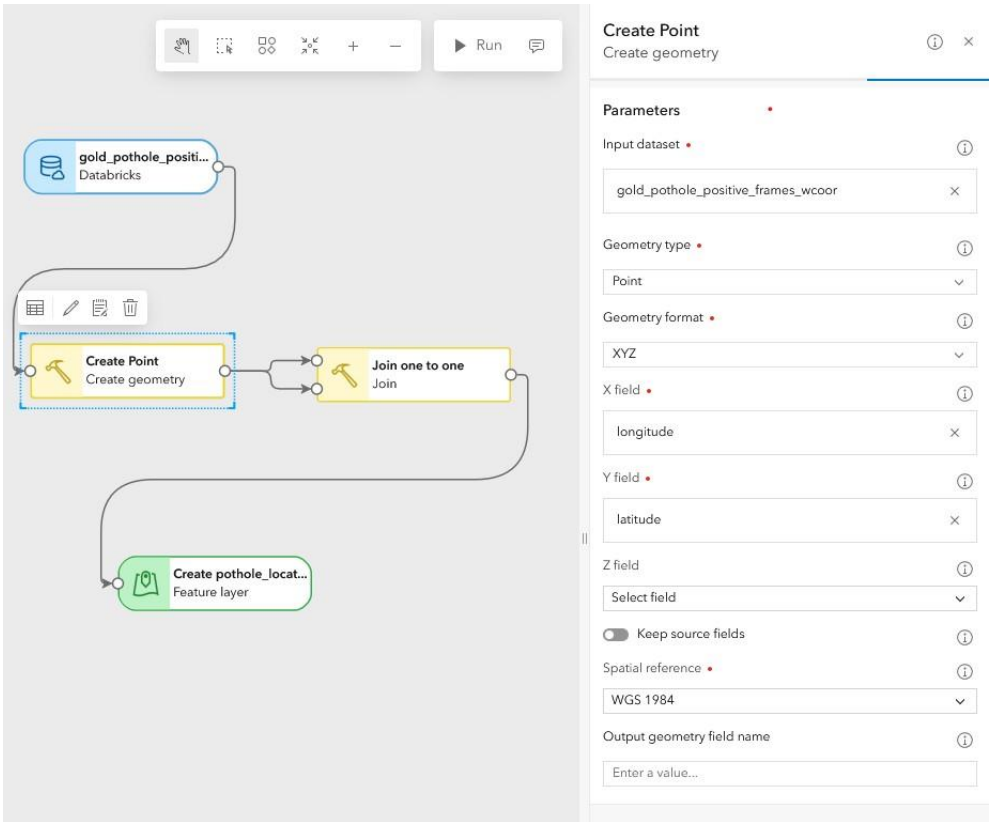
ArcGIS Data Pipelines Architecture

We can assign a schedule for this process to run. For this project, we have it scheduled on a weekly basis, similar to the prediction schedule. After the prediction table is generated, this Pipeline is scheduled to run. There is no option for setting a trigger, such as when source table is updated. So, this is merely triggered by the weekly time interval.

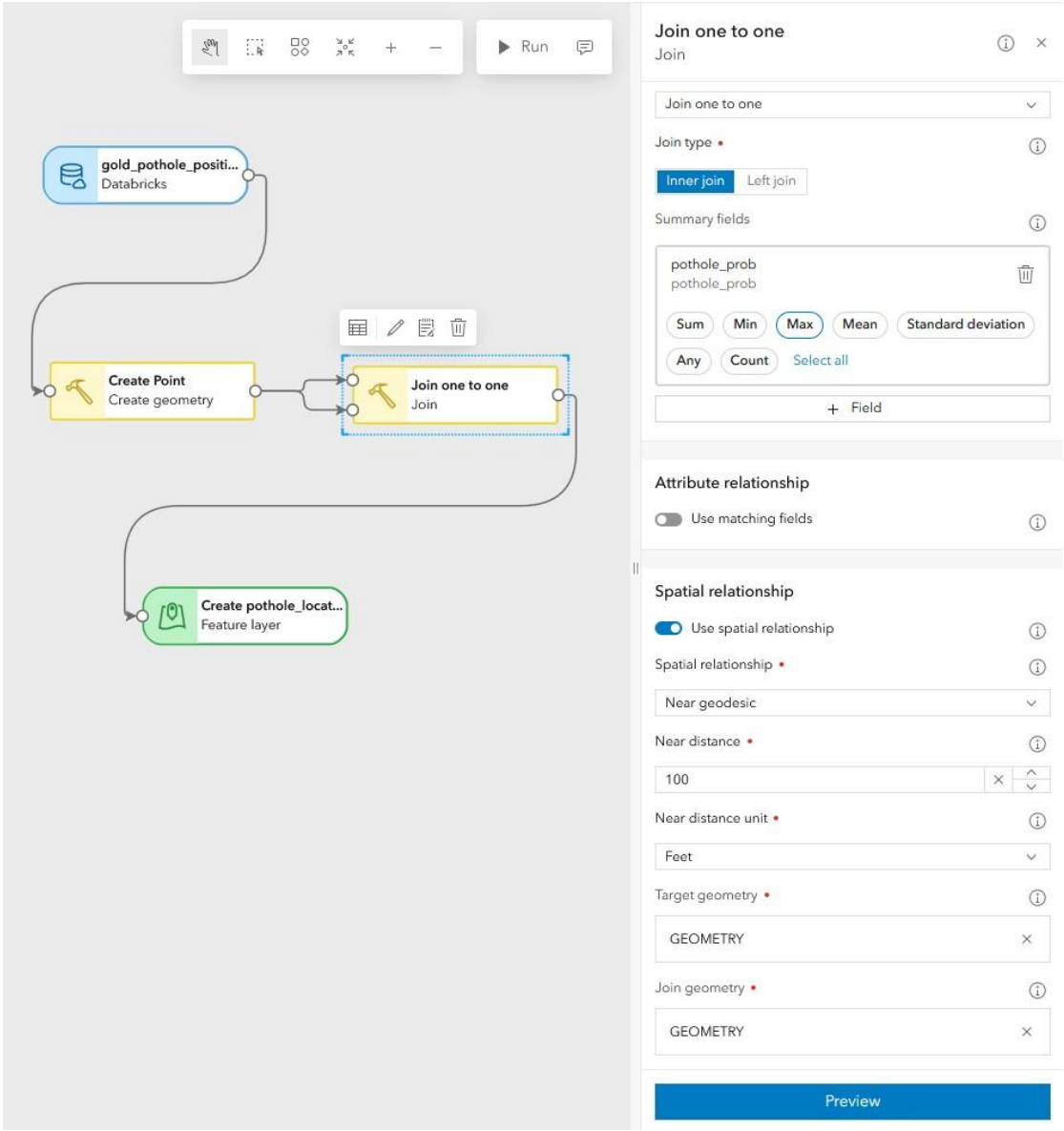
1. Reading predictions table (generated from model on unlabeled images) from Databricks catalog



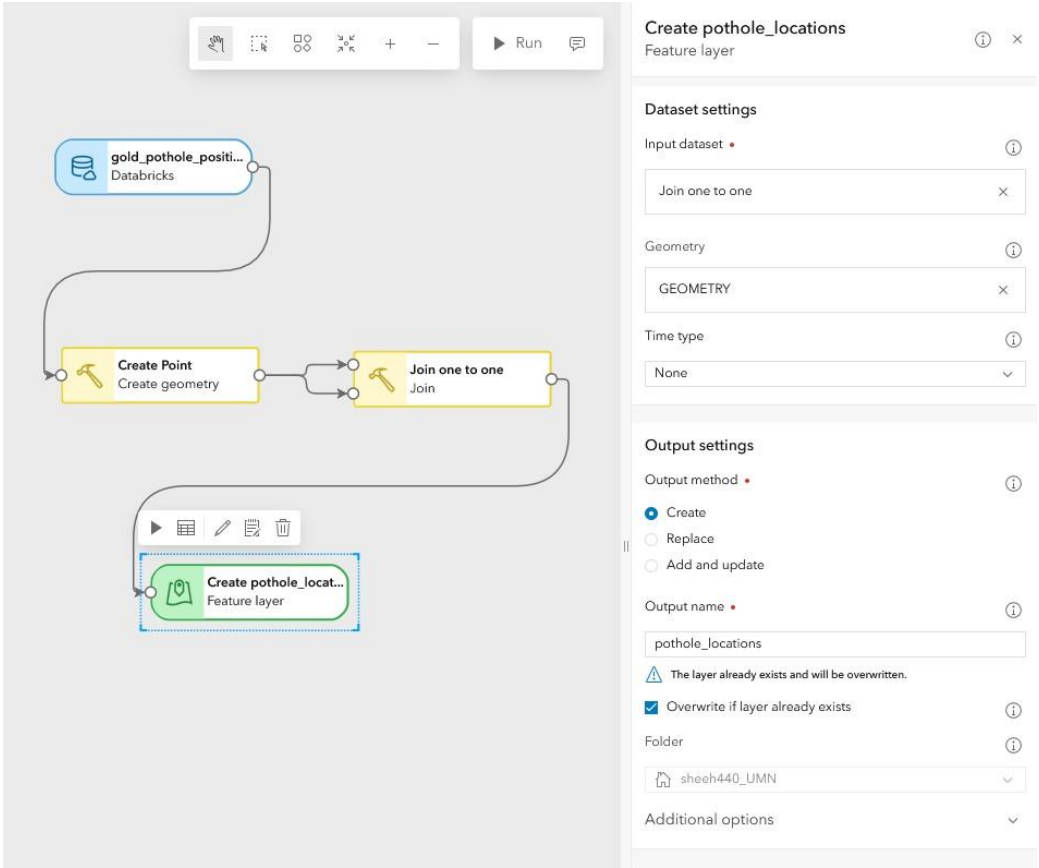
2. Creating Geometry attribute. ArcGIS requires a column with geometry information in order to map these points. For this project, these coordinates were simulated via ArcGIS Pro. Those simulated coordinates were joined to predictions table in Databricks.



3. Joining nearby points together. This makes the visualization more interpretable. There might be different images of the same pothole, and there might be lots of distinct potholes clustered together. Output row will represent the highest predicted probability of points that are joined together. There is also a column generated for the number of points the resulting point joins together, JOIN_COUNT.

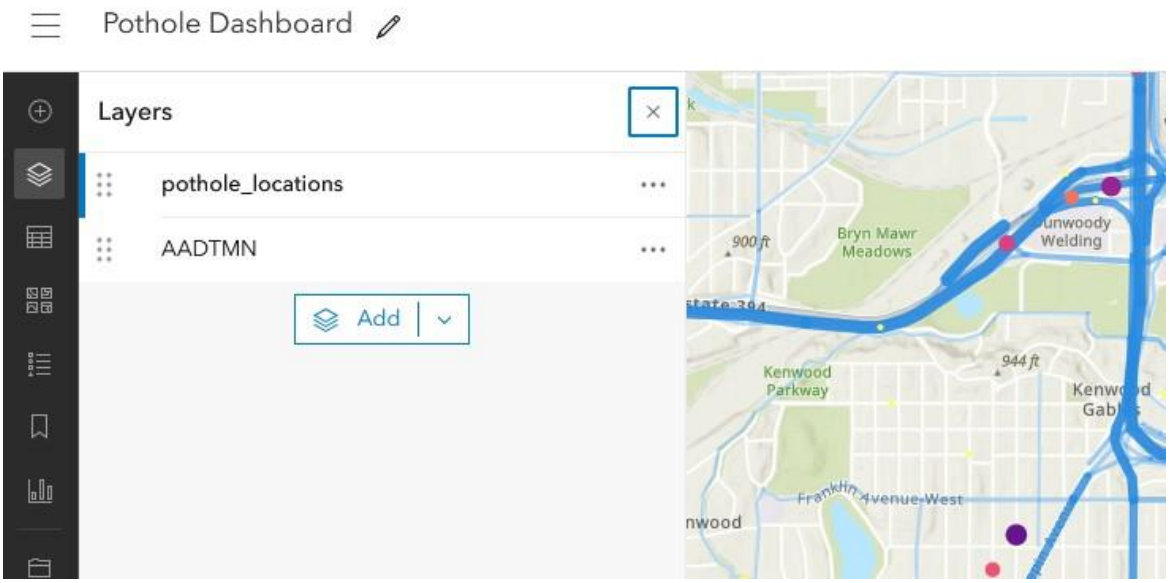


4. Exporting feature layer. This is required by ArcGIS in order to map points. This layer is configured to overwrite last week's feature layer. So, dashboard will only visualize current potholes.

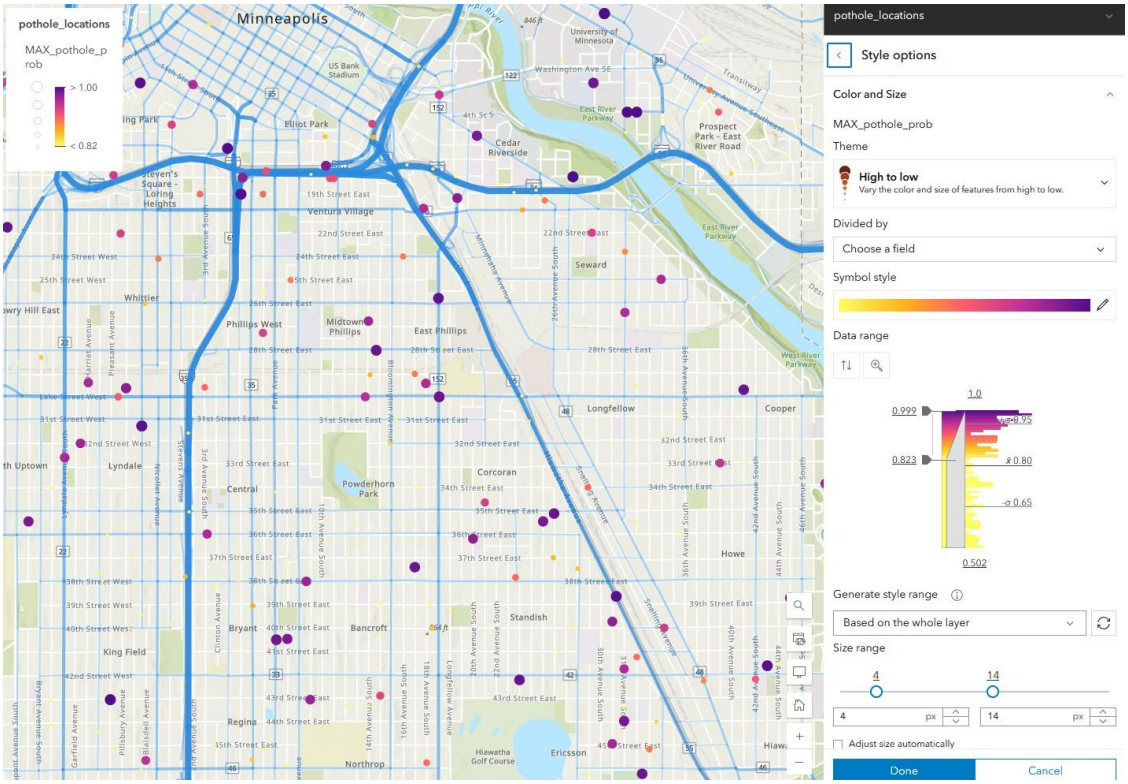


Mapping/Visualization details

5. Layers: one is the output from the Pipeline above, the other layer is the average daily traffic count on select street segments. The AADT data is publicly available.



6. Visualizing potholes. The size and color are representing the predicted probability of the image taken at those coordinates. Alternatively, color could be configured to represent count of potholes near a point. Gradient should be defined according to budget and schedule constraints.



7. **ArcGIS Web map to ArcGIS App (the final dashboard)**

   Allows for toggling visible layers and removal of points after its repaired.